Algebraic Topology

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0 Abstract Algebra Foundations

0.1 Group Theory Basics

Definition 0.1.1. Group

A **group** $\langle G, * \rangle$ is a set G, closed under a binary operaton $*$, which satisfies

- associativity of $*$
- existence of identity element e
- existence of inverse element a' of a for all $a \in G$

Definition 0.1.2. Homomorphism and Isomorphism

Let $\langle G_1, *_1 \rangle$ and $\langle G_2, *_2 \rangle$ be groups and $f : G_1 \to G_2$. We say that f is a **group isomorphism** if

- f is bijective.
- For all $a, b \in G$, $f(a *_1 b) = f(a) *_2 f(b)$.

f is a **group homomorphism** if the second condition is satisfied.

Definition 0.1.3. Permutation and Symmetric Group

Let $A \neq \varnothing$. A **permutation** of A is a bijection $\sigma \colon A \to A$. The **symmetric group** on A is

$$S_A = \mathrm{Sym}(A) = \{\sigma \colon A \to A \mid \sigma \text{ is bijective}\}$$

For $A = \{1, 2, \ldots, n\}$, denote $S_n = S_{\{1,\ldots,n\}}$ with order $|S_n| = n!$.

Definition 0.1.4. Cycle and Transposition

A k**-cycle** $(a_1\ a_2\ \cdots\ a_k) \in S_n$ $(k \geq 2)$ is defined by

$$\sigma(a_i) = a_{i+1}\ (1 \leq i < k), \quad \sigma(a_k) = a_1, \quad \sigma(x) = x\ \forall x \notin \{a_1, \ldots, a_k\}\,.$$

A 2-cycle is a **transposition**. Inverses satisfy:

$$(a_1\ a_2\ \cdots\ a_k)^{-1} = (a_k\ a_{k-1}\ \cdots\ a_1).$$

Proposition 0.1.5. Let $\mathrm{supp}(\sigma) = \{x \in \{1, \ldots, n\} \mid \sigma(x) \neq x\}$.

(a) **Disjoint Commutativity**: $\mathrm{supp}(\sigma) \cap \mathrm{supp}(\tau) = \varnothing \implies \sigma\tau = \tau\sigma$.

(b) **Disjoint Cycle Decomposition**: $\forall \pi \in S_n$, $\exists!$ partition of moved elements into disjoint cycles $c_1, \ldots, c_m$ such that

$$\pi = c_1 c_2 \cdots c_m.$$

Definition 0.1.6. General Linear Group

For $n \in \mathbb{Z}^+$,

$$\mathrm{GL}(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\} = M_n(\mathbb{R})^\times.$$

$(\mathrm{GL}(n, \mathbb{R}), \cdot)$ is nonabelian for all $n \geq 2$.

Definition 0.1.7. Dihedral Group

For $n \geq 3$, the **dihedral group** D_n is the isometry group of a regular n-gon in $\mathbb{R}^2$:

$$D_n = \langle \rho, \mu \mid \rho^n = e,\ \mu^2 = e,\ \mu\rho\mu = \rho^{-1} \rangle.$$

Theorem 0.1.8. Let $n \geq 3$. Then $|D_n| = 2n$, with canonical form:

$$D_n = \left\{ \rho^k, \mu\rho^k \mid 0 \leq k < n \right\},$$

governed by:

$$\rho^k \mu = \mu\rho^{-k} = \mu\rho^{n-k}, \quad (\rho^k)^{-1} = \rho^{n-k}, \quad (\mu\rho^k)^{-1} = \mu\rho^k.$$

Definition 0.1.9. Subgroup

Let $(G, \cdot)$ be a group. $H \subseteq G$ is a **subgroup** $(H \leq G)$ if

$$H \neq \varnothing \quad \text{and} \quad (H, \cdot|_{H \times H}) \text{ is a group.}$$

$H < G$ denotes a proper subgroup $(H \neq G)$. $H = \{e\}$ is the **trivial subgroup**; $H = G$ is the **improper subgroup**.

Theorem 0.1.10 (Subgroup Criteria)**.** Let G be a group and $H \subseteq G$.

(a) **Three-Step Test**:

$$H \leq G \iff (\forall a, b \in H \implies ab \in H) \wedge (e \in H) \wedge (\forall a \in H \implies a^{-1} \in H).$$

(b) **One-Step Test**:

$$H \leq G \iff H \neq \varnothing \wedge (\forall a, b \in H \implies ab^{-1} \in H).$$

(c) **Finite Subgroup Test**: If $|H| < \infty$ and $H \neq \varnothing$,

$$H \leq G \iff (\forall a, b \in H \implies ab \in H).$$

Example 0.1.11 (Standard Subgroups)**.**

(a) **Center**: $Z(G) = \{z \in G \mid zg = gz\ \forall g \in G\} \leq G$ is abelian.
(b) **Special Linear Group**: $\mathrm{SL}(n, \mathbb{R}) = \{A \mid \det(A) = 1\} \leq \mathrm{GL}(n, \mathbb{R})$.
(c) **Roots of Unity**: $U_n = \{z \in \mathbb{C} \mid z^n = 1\} \leq U = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$.

Definition 0.1.12. Cyclic Subgroup and Generator

For $a \in G$, the **cyclic subgroup generated by** a is

$$\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\} \leq G \quad \text{(in additive notation, } \{na \mid n \in \mathbb{Z}\}\text{)}.$$

G is **cyclic** if $\exists a \in G$ such that $G = \langle a \rangle$; a is called a **generator** of G.

Theorem 0.1.13. $G = \langle a \rangle \implies G$ is abelian.

Theorem 0.1.14 (Division Algorithm for $\mathbb{Z}$)**.**

$$\forall m \in \mathbb{Z}^+, \forall n \in \mathbb{Z}, \quad \exists!\, q, r \in \mathbb{Z} \text{ such that } n = mq + r \quad \text{and} \quad 0 \leq r < m.$$

Theorem 0.1.15 (Subgroups of Cyclic Groups)**.** Let $G = \langle a \rangle$.

(a) $H \leq G \implies H$ is cyclic.
(b) The subgroups of $(\mathbb{Z}, +)$ are precisely $n\mathbb{Z} = \langle n \rangle$ for $n \in \mathbb{Z}_{\geq 0}$.

Definition 0.1.16. Greatest Common Divisor

For $r \in \mathbb{Z}^+, s \in \mathbb{Z}_{\geq 0}$, $\gcd(r, s)$ is the unique positive generator of $r\mathbb{Z} + s\mathbb{Z}$:

$$\langle \gcd(r,s) \rangle = \{nr + ms \mid n, m \in \mathbb{Z}\} \implies \exists n, m \in \mathbb{Z} \colon \gcd(r,s) = nr + ms.$$

Theorem 0.1.17 (Classification of Cyclic Groups)**.** Let $G = \langle a \rangle$.

(a) $|G| = \infty \implies G \cong (\mathbb{Z}, +)$ via $a^k \mapsto k$.
(b) $|G| = n < \infty \implies G \cong (\mathbb{Z}_n, +_n)$ via $a^k \mapsto k \pmod{n}$.

Theorem 0.1.18 (Orders, Generators, and Subgroup Lattice)**.** Let $G = \langle a \rangle$ with $|G| = n$, and let $b = a^s \in G$.

(a) **Element Order**:

$$|a^s| = |\langle a^s \rangle| = \frac{n}{\gcd(n, s)}, \quad \text{and} \quad \langle a^s \rangle = \langle a^{\gcd(n,s)} \rangle.$$

(b) **Generators**:

$$\langle a^s \rangle = G \iff \gcd(n, s) = 1.$$

The number of distinct generators of G is $\phi(n) = |\{k \in \{1, \ldots, n\} \mid \gcd(n, k) = 1\}|$.

(c) **Lagrange Divisibility**: $H \leq G \implies |H| \mid n$.

(d) **Subgroup Uniqueness**: For each divisor $d \mid n$ $(d > 0)$, there exists a **unique** subgroup of order d:

$$H_d = \langle a^{n/d} \rangle = \left\{x \in G \mid x^d = e\right\}, \quad |H_d| = d.$$

Theorem 0.1.19 (Subgroup Intersections)**.** Let $\{H_i\}_{i\in I}$ be any nonempty collection of subgroups of G. Then

$$\bigcap_{i\in I} H_i \leq G.$$

Definition 0.1.20. Subgroup Generated by a Subset

For $\varnothing \neq S \subseteq G$:

$$\langle S\rangle = \bigcap \{H \leq G \mid S \subseteq H\}.$$

S is a **generating set** of G if $\langle S\rangle = G$. If $\exists S \subseteq G$ with $|S| < \infty$ and $\langle S\rangle = G$, G is **finitely generated**.

Theorem 0.1.21 (Constructive Characterization)**.** For $\varnothing \neq S \subseteq G$:

$$\langle S\rangle = \{x_1^{\epsilon_1} x_2^{\epsilon_2} \cdots x_k^{\epsilon_k} \mid k \in \mathbb{N},\, x_j \in S,\, \epsilon_j \in \{\pm 1\}\}.$$

Definition 0.1.22. Cayley Digraph

Let G be a finite group and $S \subseteq G$ a generating set. The **Cayley digraph** $\Gamma(G,S) = (V,E)$ is defined by:

- $V = G$;
- $E = \{(g, gs; s) \mid g \in G,\, s \in S\}$, representing a directed arc $g \xrightarrow{s} gs$.

When $s = s^{-1}$, (g, gs) and (gs, g) collapse to an undirected edge.

Proposition 0.1.23 (Structural Properties of $\Gamma(G,S)$)**.**

(a) **Connectivity**: $\Gamma(G,S)$ is connected $\iff \langle S\rangle = G$.
(b) **Valency**: $\forall g \in G$, $\forall s \in S$, $\deg_{\text{in}}^{(s)}(g) = \deg_{\text{out}}^{(s)}(g) = 1$.
(c) **Vertex Transitivity**: $G \leq \text{Aut}(\Gamma(G,S))$ via left regular action $\lambda_h(g) = hg$.
(d) **Cycles and Relations**: Directed closed paths from e:

$$e \xrightarrow{s_1^{\epsilon_1}} \cdots \xrightarrow{s_k^{\epsilon_k}} e \iff s_1^{\epsilon_1} \cdots s_k^{\epsilon_k} = e \text{ in } G.$$

Example 0.1.24. Below is a Cayley digraph of D_4 with generating set $S = \{\rho, \mu\}$

Exercise 0.1.25. Let G be an arbitrary group and let $a, b \in G$.

- Prove that if $|ab| = n < \infty$, then $|ba| = n$.

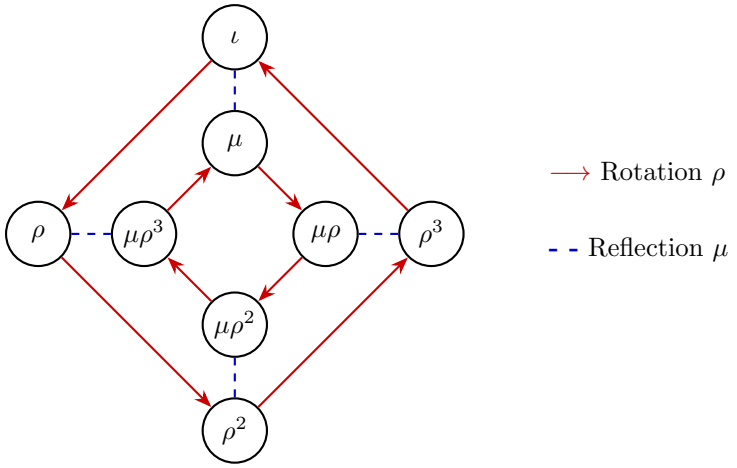
  Proof. Notice

  $$(ab)^n = a(ba)^{n-1}b = e \implies a(ba)^n = ea = a \implies (ba)^n = e \implies |ba| \leq n$$

  and

  $$((ba)^m = e \implies (ab)^m = e) \implies |ba| \geq |ab| = n$$

  Then $|ba| = n$. □



- Deduce that for any $g \in G$ and $x \in G$, $|x| = |gxg^{-1}|$

  Proof. We have

  $$(gxg^{-1})^m = gx^m g^{-1} = e \implies gx^m = g \implies x^m = e \implies \left|gxg^{-1}\right| \geq |x|$$

  We also have

  $$x^m = e \implies gx^m g^{-1} = e \implies gxg^{-1})^m = e \implies |x| \geq \left|gxg^{-1}\right|$$

  Therefore, $|x| = |gxg^{-1}|$. □

  Remark. In fact, let $a = gx$ and $b = g^{-1}$ then $|ab| = |gxg^{-1}| = |ba| = |x|$.

Exercise 0.1.26. Let H and K be subgroups of a group G, and define the subset $HK = \{hk : h \in H, k \in K\}$

- Prove that if G is abelian, then $HK \leq G$.

  Proof. Let $hk, h'k' \in HK$, then we have

  $$(hk)(h'k')^{-1} = hkk'^{-1}h'^{-1} = (hh'^{-1})(kk'^{-1}) \in HK$$

  implying that $HK \leq G$. □

- Give a counterexample when G is nonabelian.

  Proof. Let $G = D_3, H = \langle\mu\rangle, K = \langle\mu\rho\rangle$, then $HK = \{e, \mu, \rho, \mu\rho\}$ is not a subgroup of G. □

Exercise 0.1.27. Let G be a finite cyclic group. Prove that if $H, K \leq G$ with $H \neq K$, then $|H| \neq |K|$.

Proof. Let $G = \langle a\rangle$ and $|G| = n$. Then $H, K \leq G$ implies $H = \langle a^p\rangle = \langle a^{\gcd(n,p)}\rangle, K = \langle a^q\rangle = \langle a^{\gcd(n,q)}\rangle$ for some $p, q \in \mathbb{Z}^+$, then

$$H \neq K \implies \gcd(n,p) \neq \gcd(n,q) \implies |H| = \frac{n}{\gcd(n,p)} \neq \frac{n}{\gcd(n,q)} = |K|$$

□

Exercise 0.1.28. Prove that a group G with $|G| > 1$ has no proper nontrivial subgroups if and only if G is finite and is isomorphic to $\mathbb{Z}_p$ for some prime p.

Proof. $\implies$: Let $a \neq e \in G$.. Then $\langle a \rangle \leq G$ must equals G for otherwise it will be a proper nontrivial subgroup of G. Then G is cyclic. If G is infinite, then $\langle a \rangle = \langle a^2 \rangle = G$ leads to a contradiction since $a \notin \langle a^2 \rangle$. If $|G| = mn$ for some $m, n > 1$, then $|a^m| = mn/\gcd(mn, m) = n$ implying that $\langle a^m \rangle$ is a proper nontrivial subgroup of G. Therefore, $G = \langle a \rangle$ and $|G| = p$ with p prime. It follows that $G \cong \mathbb{Z}_p$.

$\impliedby$: Let $G \cong \mathbb{Z}_p$, then $G = \langle a \rangle$ and $|G| = p$. If $H \leq G$, then $H = \langle a^s \rangle = \langle a^{\gcd(p,s)} \rangle$. Since p is prime, we know $H = \{e\}$ or $H = G$. □

0.2 Finitely Generated Abelian Groups

Definition 0.2.1. Direct Product of Groups

Let $G_1, G_2, \ldots, G_n$ be groups. The **direct product** $\prod_{i=1}^n G_i = G_1 \times G_2 \times \cdots \times G_n$ is the Cartesian product endowed with componentwise binary operation:

$$(a_1, a_2, \ldots, a_n)(b_1, b_2, \ldots, b_n) = (a_1 b_1, a_2 b_2, \ldots, a_n b_n).$$

The identity element is $(e_1, e_2, \ldots, e_n)$, and $(a_1, \ldots, a_n)^{-1} = (a_1^{-1}, \ldots, a_n^{-1})$. It therefore forms a group.

Theorem 0.2.2. The group $\prod_{i=1}^n \mathbb{Z}_{m_i}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1 m_2 \cdots m_n}$ if and only if $\gcd(m_i, m_j) = 1$ for all $i \neq j$.

Theorem 0.2.3. Let $(a_1, a_2, \ldots, a_n) \in \prod_{i=1}^n G_i$. If each a_i has finite order r_i in G_i, then

$$|(a_1, a_2, \ldots, a_n)| = \operatorname{lcm}(r_1, r_2, \ldots, r_n).$$

Theorem 0.2.4 (**Fundamental Theorem of Finitely Generated Abelian Groups - Primary Factor Form**)**.** Every finitely generated abelian group G is isomorphic to a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}} \times \underbrace{\mathbb{Z} \times \cdots \times \mathbb{Z}}_{\beta \text{ copies}},$$

where p_i are primes (not necessarily distinct) and $r_i \in \mathbb{Z}^+$. β is called the **Betti number** of G.

Theorem 0.2.5 (**Fundamental Theorem of Finitely Generated Abelian Groups - Invariant Factor Form**)**.** Every finitely generated abelian group G is isomorphic to a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_k} \times \underbrace{\mathbb{Z} \times \cdots \times \mathbb{Z}}_{\beta \text{ copies}},$$

where each $d_i \geq 2$ and $d_1 \mid d_2 \mid \cdots \mid d_k$. The **invariant factors** $d_1, \ldots, d_k$ and the Betti number β are uniquely determined.

Definition 0.2.6. Decomposable and Indecomposable Groups

A group G is **decomposable** if it is isomorphic to the direct product of two proper nontrivial subgroups: $G \cong H \times K$ with $H, K \neq \{e\}$. Otherwise, G is **indecomposable**.

Theorem 0.2.7.

$$G \text{ is finite indecomposable abelian} \iff G \cong \mathbb{Z}_{p^r} \text{ for some prime } p \text{ and } r \in \mathbb{Z}^+.$$

Theorem 0.2.8. If G is a finite abelian group and $m \mid |G|$, then G contains a subgroup of order m.

Theorem 0.2.9. If $m \in \mathbb{Z}^+$ is square-free, then every abelian group of order m is cyclic.

Exercise 0.2.10. Let $G = \mathbb{Z}_4 \times \mathbb{Z}_{18} \times \mathbb{Z}_{15}$ and $H = \mathbb{Z}_3 \times \mathbb{Z}_{36} \times \mathbb{Z}_{10}$. Are they isomorphic?

Proof. We have

$$G \cong \mathbb{Z}_4 \times (\mathbb{Z}_2 \times \mathbb{Z}_9) \times (\mathbb{Z}_3 \times \mathbb{Z}_5)$$

and

$$H \cong \mathbb{Z}_3 \times (\mathbb{Z}_4 \times \mathbb{Z}_9) \times (\mathbb{Z}_2 \times \mathbb{Z}_5)$$

It follows that $G \cong H$. □

Exercise 0.2.11. Let G be a finite abelian group. Prove that G is not cyclic if and only if G contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ for some prime p.

Proof. Since G is finite abelian, then

$$G \cong \prod_{i=1}^{n} \mathbb{Z}_{(p_i)^{r_i}}$$

By Theorem 0.2.2, if it is not cyclic, then $\gcd((p_i)^{r_i}, (p_j)^{r_j}) > 1$ for some $i \neq j$, implying that $p_i = p_j$ and then

$$\mathbb{Z}_{p_i} \times \mathbb{Z}_{p_i} \cong \langle (p_i)^{r_i - 1} \rangle \times \langle (p_j)^{r_j - 1} \rangle \leq \mathbb{Z}_{(p_i)^{r_i}} \times \mathbb{Z}_{(p_j)^{r_j}} \leq \prod_{i=1}^{n} \mathbb{Z}_{(p_i)^{r_i}} \cong G$$

If G is cylic, then $p_i \neq p_j$ for all $i \neq j$. It follows that if $H \leq G$, then

$$H \cong \prod_{i=1}^{n} Z_{(p_i)^{q_i}}, q_i \leq r_i$$

which is not isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ since p_i are mutually distinct. □

Remark. In fact, when G is cylic, we can deduce that $H \leq G$ is also cyclic, which directly tells proves the result.

Remark. Generally, one can prove that:

> Let G be a finite abelian group, and let $x_1, \ldots, x_n \geq 2$ be integers. The group $H = \prod_{i=1}^{n} \mathbb{Z}_{x_i}$ is isomorphic to a subgroup of G if and only if G can be decomposed as a direct product of cyclic groups $G \cong \prod_{j=1}^{m} \mathbb{Z}_{y_j}$ such that $m \geq n$ and $x_i \mid y_i$ for all $i \in \{1, \ldots, n\}$.

Exercise 0.2.12. Let G, H, K be finitely generated abelian groups. Show that

$$G \times K \cong H \times K \implies G \cong H$$

Proof. Since G, H, K are finitely generated abelian groups, so are $G \times K$ and $H \times K$. Since $G \times K \cong H \times K$, so they have the same Betti number and the same collection of $(p_i)^{r_i}$ after decomposition. It follows that after picking up the part given by K, we know G and H have the same Betti number and the same collection of $(p_i)^{r_i}$ after decomposition, which implies $G \cong H$. □

0.3 Cosets and the Theorem of Lagrange

Theorem 0.3.1. Let $H \leq G$. The relation $\sim_L$ defined on G by

$$a \sim_L b \iff a^{-1}b \in H$$

is an equivalence relation on G. Similarly, $a \sim_R b \iff ab^{-1} \in H$ is an equivalence relation.

Definition 0.3.2. Coset

Let $H \leq G$ and $a \in G$.

- The **left coset** of H containing a is the equivalence class under $\sim_L$:

$$aH = \{ah \mid h \in H\}.$$

- The **right coset** of H containing a is the equivalence class under $\sim_R$:

$$Ha = \{ha \mid h \in H\}.$$

Theorem 0.3.3. Let $H \leq G$. For every $a \in G$, the map $f \colon H \to aH$ defined by $f(h) = ah$ is a bijection. Consequently,

$$|aH| = |H| = |Ha| \quad (\forall a \in G).$$

Theorem 0.3.4 (Theorem of Lagrange)**.** Let H be a subgroup of a finite group G. Then the order of H divides the order of G:

$$|H| \mid |G|.$$

Corollary 0.3.5. Every group of prime order p is cyclic, hence isomorphic to $\mathbb{Z}_p$.

Theorem 0.3.6. Let G be a finite group. For every $g \in G$, $|g|$ divides $|G|$. In particular, $g^{|G|} = e$.

Definition 0.3.7. Index of a Subgroup

Let $H \leq G$. The **index** of H in G, denoted $(G : H)$ or $[G : H]$, is the cardinality of the set of left (or right) cosets of H in G. When G is finite,

$$[G : H] = \frac{|G|}{|H|}.$$

Theorem 0.3.8 (Tower Law for Subgroup Indices)**.** Let $K \leq H \leq G$. If $[G : H]$ and $[H : K]$ are finite, then $[G : K]$ is finite and

$$[G : K] = [G : H] \cdot [H : K].$$

Theorem 0.3.9 (Cosets of Homomorphism Kernels)**.** Let $\phi\colon G \to G'$ be a group homomorphism. Then

(a) Left and right cosets of $\ker(\phi)$ coincide:

$$a\ker(\phi) = \ker(\phi)a \quad (\forall a \in G).$$

(b) Two elements $a, b \in G$ belong to the same coset of $\ker(\phi)$ if and only if $\phi(a) = \phi(b)$:

$$a\ker(\phi) = b\ker(\phi) \iff \phi(a) = \phi(b).$$

Corollary 0.3.10. A group homomorphism $\phi\colon G \to G'$ is injective if and only if

$$\ker(\phi) = \{e_G\}.$$

Exercise 0.3.11. Let $\sigma = (1, 2, 5, 4)(2, 3)$ in S_5, find the index of $\langle\sigma\rangle$ in S_5.

Proof. Notice $\sigma = (1, 2, 3, 5, 4)$, then the order of σ is 5 in S_5, implying the index given by

$$[S_5 : \langle\sigma\rangle] = \frac{|S_5|}{|\langle\sigma\rangle|} = \frac{5!}{5} = 24$$

□

Exercise 0.3.12. Let H be a subgroup of a group G, let $a, b \in G$. Is it true that

$$aH = bH \implies Ha^{-1} = Hb^{-1}$$

Proof. The statement is true. Since $aH = bH$, then $ah_a = bh_b$ for some $h_a, h_b \in H$, it follows that $b^{-1}a \in H$. Therefore, for $hb^{-1} \in Hb^{-1}$, we have $hb^{-1} = hb^{-1}aa^{-1} = (hb^{-1}a)a^{-1} \in Ha^{-1}$. Then $Hb^{-1} \subset Ha^{-1}$, similarly we can deduce that $H^{-1} \subset Hb^{-1}$, thus $Ha^{-1} = Hb^{-1}$. □

Exercise 0.3.13. Let G be a group of order pq, where p and q are prime numbers. Show that every proper subgroup of G is cyclic.

Proof. By Theorem of Lagrange we know every proper subgroup H has order $|H| = n \mid pq$ implying $n = 1$ or $n = p$ or $n = q$. If $n = 1$ then H is trivially cyclic, if $n = p$ or $n = q$ then Corollary 0.3.5 tells that H is cyclic. □

0.4 Normal Subgroups and Factor Groups

Definition 0.4.1. Normal Subgroup

A subgroup $N \leq G$ is **normal** in G, written $N \trianglelefteq G$, if

$$gN = Ng \qquad (\forall g \in G)$$

Theorem 0.4.2 (Characterizations of Normality)**.** Let $N \leq G$. The following are equivalent:

(a) $N \trianglelefteq G$.
(b) $gNg^{-1} = N$ for every $g \in G$.
(c) $gng^{-1} \in N$ for every $g \in G$ and $n \in N$.
(d) There is a group homomorphism $\phi : G \to G'$ such that $\ker(\phi) = N$.

Corollary 0.4.3. Every subgroup of an abelian group is normal.

Definition 0.4.4. Factor Group

Let $N \trianglelefteq G$. The **factor group**, or **quotient group**, of G by N is

$$G/N = \{gN \mid g \in G\}, \qquad (aN)(bN) = (ab)N.$$

Its identity element is N, and

$$(gN)^{-1} = g^{-1}N.$$

Definition 0.4.5. Canonical Projection

For $N \trianglelefteq G$, the map

$$q\colon G \to G/N, \qquad q(g) = gN,$$

is called the **canonical projection**.

Theorem 0.4.6. The canonical projection $q\colon G \to G/N$ is a surjective homomorphism with

$$\ker(q) = N.$$

Theorem 0.4.7 (First Isomorphism Theorem)**.** Let $\phi\colon G \to G'$ be a group homomorphism, and set $K = \ker(\phi)$. Then

$$\bar{\phi}\colon G/K \to \mathrm{im}(\phi), \qquad \bar{\phi}(gK) = \phi(g),$$

is an isomorphism. Thus

$$\boxed{G/\ker(\phi) \cong \mathrm{im}(\phi).}$$

If $q\colon G \to G/K$ is the canonical projection, then $\bar{\phi}$ is the unique map satisfying

$$\bar{\phi}(q(g)) = \phi(g) \qquad (\forall g \in G).$$

In particular, if ϕ is surjective, then

$$G/\ker(\phi) \cong G'.$$

Remark. To identify a quotient G/N, it is often useful to construct a surjective homomorphism $\phi\colon G \to H$ with $\ker(\phi) = N$. The First Isomorphism Theorem then gives $G/N \cong H$.

Definition 0.4.8. Automorphism and Inner Automorphism

An **automorphism** of G is an isomorphism $G \to G$. For each $g \in G$, the map

$$i_g\colon G \to G, \qquad i_g(x) = gxg^{-1},$$

is an automorphism, called the **inner automorphism**· induced by g. The operation $x \mapsto gxg^{-1}$ is called **conjugation** by g.

Proposition 0.4.9. A subgroup $N \leq G$ is normal if and only if it is invariant under every inner automorphism:

$$i_g(N) = N \qquad (\forall g \in G).$$

Remark. Normality means that conjugation preserves N as a set. It does not require $gng^{-1} = n$ for every $g \in G$ and $n \in N$.

Exercise 0.4.10. Let F be the group of all functions $\mathbb{R} \to \mathbb{R}$, under pointwise addition:

$$(f + g)(x) = f(x) + g(x)$$

Fix $c \in \mathbb{R}$, and define

$$\phi_c : F \to \mathbb{R}, \qquad \phi_c(f) = f(c).$$

- Prove that ϕ_c is a group homomorphism.

  Proof. ϕ_c is a homomorphism since for $f, g \in F$, we have

  $$\phi_c(f + g) = (f + g)(c) = f(c) + g(c) = \phi_c(f) + \phi_c(g)$$

  □

- Determine $\ker(\phi_c)$.

  Proof.
  $$\ker(\phi_c) = \{f \in F \mid \phi_c(f) = 0\} = \{f : \mathbb{R} \to \mathbb{R} \mid f(c) = 0\}$$
  □

- Describe the coset of $\ker(\phi_c)$ containing the constant function $f(x) = 1$.

  Proof. The coset is given by
  $$f + \ker(\phi_c) = 1 + \{f : \mathbb{R} \to \mathbb{R} \mid f(c) = 0\} = \{f : \mathbb{R} \to \mathbb{R} \mid f(c) = 1\}$$
  □

- Identify a familiar group isomorphic to $F/\ker(\phi_c)$.

  Proof. By First Isomorphism Theorem, since $\phi_c : F \to \mathbb{R}$ is a group homomorphism, then
  $$F/\ker(\phi_c) \cong \text{im}(\phi_c) = \mathbb{R}$$
  □

Exercise 0.4.11. Let $H, N \leq G$ with $N \trianglelefteq G$. Prove that $H \cap N \trianglelefteq H$ and give a example showing that $H \cap N$ need not be normal in G.

Proof. Let $x \in H, y \in H \cap N$, then $xyx^{-1} \in H \cap N$ since
$$x \in G, y \in N \implies xyx^{-1} \in N$$
$$x, y \in H \implies xyx^{-1} \in H$$
Let $G = D_3, N = G, H = \langle \mu \rangle$, then $N \cap H = \langle \mu \rangle$ is not a normal subgroup of G since $\mu\rho \neq \rho\mu$. □

Exercise 0.4.12. Let G, G' be groups with $H \trianglelefteq G$ and $H' \trianglelefteq G'$. Let $\phi : G \to G'$ be a homomorphism satisfying:
$$\phi(H) \subset H'$$
Prove that ϕ induces a well-defined homomorphism
$$\phi_* : G/H \to G'/H' \qquad \phi_*(gH) = \phi(g)H'$$

Proof. If $xH = yH$, then
$$xy^{-1} \in H \implies \phi(xy^{-1}) \in H' \implies \phi(x)H' = \phi(y)H' \implies \phi_*(xH) = \phi_*(yH)$$
which implies ϕ_* is well-defined. Since ϕ is a homomorphism, then
$$\phi_*(xyH) = \phi(xy)H' = (\phi(x)H')(\phi(y)H') = \phi_*(xH)\phi_*(yH)$$
which implies ϕ_* is a homomorphism □

0.5 Factor-Group Computations and Simple Groups

Theorem 0.5.1 (Quotients of Direct Products)**.** Let H and K be groups. Then

$$H \times \{e_K\} \trianglelefteq H \times K, \qquad \{e_H\} \times K \trianglelefteq H \times K,$$

and the coordinate projections induce isomorphisms

$$\frac{H \times K}{H \times \{e_K\}} \cong K, \qquad \frac{H \times K}{\{e_H\} \times K} \cong H.$$

Theorem 0.5.2 (Cyclic and Abelian Quotients)**.** Let $N \trianglelefteq G$.

(a) If G is cyclic, then G/N is cyclic. More precisely,

$$G = \langle a \rangle \implies G/N = \langle aN \rangle.$$

(b) If G is abelian, then G/N is abelian.

Example 0.5.3. For every group G and every positive integer n,

$$G/\{e_G\} \cong G, \qquad G/G \cong \{e\}, \qquad \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n.$$

Theorem 0.5.4 (Images and Preimages of Normal Subgroups)**.** Let $\phi\colon G \to G'$ be a group homomorphism.

(a) If $N \trianglelefteq G$, then
$$\phi(N) \trianglelefteq \phi(G).$$

(b) If $M \trianglelefteq \phi(G)$, then
$$\phi^{-1}(M) \trianglelefteq G.$$

(c) In particular, if $M \trianglelefteq G'$, then $\phi^{-1}(M) \trianglelefteq G$.

Remark. The image of a normal subgroup is normal in $\phi(G)$, but need not be normal in the whole codomain G'.

Definition 0.5.5. Simple Group

A group G is **simple** if $G \neq \{e\}$ and its only normal subgroups are $\{e\}$ and G.

Theorem 0.5.6. An abelian group is simple if and only if it is isomorphic to $\mathbb{Z}_p$ for some prime p.

Theorem 0.5.7 (Simplicity of Alternating Groups)**.** The alternating group A_n is simple for every $n \geq 5$.

Definition 0.5.8. Maximal Normal Subgroup

A **maximal normal subgroup** of G is a proper normal subgroup $M \trianglelefteq G$ such that there is no normal subgroup $N \trianglelefteq G$ satisfying

$$M < N < G.$$

Theorem 0.5.9. Let $M \trianglelefteq G$. Then

$$\boxed{M \text{ is maximal normal in } G \iff G/M \text{ is simple.}}$$

Remark. A maximal normal subgroup is maximal among proper *normal* subgroups. It need not be maximal among all proper subgroups.

Definition 0.5.10. Center of a Group

The **center** of G is

$$Z(G) = \{z \in G \mid zg = gz \text{ for every } g \in G\}.$$

Proposition 0.5.11. The center $Z(G)$ is an abelian normal subgroup of G. Moreover,

$$G \text{ is abelian} \iff Z(G) = G.$$

Definition 0.5.12. Commutator and Commutator Subgroup

For $a, b \in G$, the element

$$[a, b] = aba^{-1}b^{-1}$$

is called their **commutator**. The **commutator subgroup** of G is

$$[G, G] = \langle [a, b] \mid a, b \in G \rangle.$$

Remark. The commutator subgroup is generated by all commutators. The set of individual commutators need not itself be a subgroup.

Theorem 0.5.13. The commutator subgroup satisfies $[G, G] \trianglelefteq G$. For every $N \trianglelefteq G$,

$$\boxed{G/N \text{ is abelian} \iff [G, G] \leq N.}$$

Thus $[G, G]$ is the smallest normal subgroup of G whose quotient is abelian.

Definition 0.5.14. Abelianization

The **abelianization** of G is the abelian group

$$G_{\text{ab}} = G/[G, G].$$

Corollary 0.5.15 (Universal Property of Abelianization)**.** Let A be an abelian group and let $\phi\colon G \to A$ be a homomorphism. There is a unique homomorphism

$$\bar{\phi}\colon G_{\text{ab}} \to A$$

such that $\phi = \bar{\phi} \circ q$, where $q\colon G \to G_{\text{ab}}$ is the canonical projection. Explicitly,

$$\bar{\phi}(g[G,G]) = \phi(g).$$

Exercise 0.5.16. Classify the group

$$\frac{\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}}{\langle(3,3,3)\rangle}$$

according to the Fundamental Theorem of Finitely Generated Abelian Groups.

Proof. Define $\phi : \mathbb{Z}\times\mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}_3\times\mathbb{Z}\times\mathbb{Z}$ by $\phi(x,y,z) = ((x+y-z) \mod 3, x-y, y-z)$ be a homormophism with $\ker(\phi) = \langle(3,3,3)\rangle$. It follows that

$$\frac{\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}}{\langle(3,3,3)\rangle} \cong \text{im}(\phi) = \mathbb{Z}_3\times\mathbb{Z}\times\mathbb{Z}$$

□

Exercise 0.5.17. Find both the center $Z(D_4)$ and the commutator subgroup $[D_4, D_4]$ of the group D_4.

Proof. One can varify that $Z(D_4) = \{e, \rho^2\}$.

Let $N = \langle\rho^2\rangle \trianglelefteq D_4$. Notice for $a^2 \in N$ for all $a \in D_4$, implying that $aN = a^{-1}N$. It follows that for $aN, bN \in D_4/N$,

$$abN = a^{-1}b^{-1}N = (ba)^{-1}N = baN$$

which is abelian. Hence $[D_4, D_4] \subset N$. Moreover, $\mu\rho(\mu)^{-1}(\rho)^{-1} = \rho^2$ tells $[D_4, D_4] = N = \langle\rho^2\rangle$. □

Exercise 0.5.18. Prove that if G is nonabelian, then

$$G/Z(G)$$

is not cyclic

Proof. Assume $G/Z(G) = \langle gZ(G)\rangle$ is cyclic. Let $a = g^n z_a, b = g^m z_b$ where $z_a, z_b \in Z(G)$. It follows that $ab = g^n z_a g^m z_b = g^m z_b g^n z_a = ba$ since g^n, g^m, z_a, z_b all commutes with each other, which contradicts to G being nonabelian. Therefore, $G/Z(G)$ is not cyclic. □

0.6 Isomorphism Theorems

Lemma 0.6.1 (Normal-Subgroup Correspondence)**.** Let $N \trianglelefteq G$, and let $q\colon G \to G/N$ be the canonical projection. There is an inclusion-preserving bijection

$$\{L \trianglelefteq G \mid N \leq L\} \longleftrightarrow \{M \trianglelefteq G/N\}$$

given by

$$L \longmapsto L/N, \qquad M \longmapsto q^{-1}(M).$$

These assignments are mutually inverse.

Definition 0.6.2. Product and Join of Subgroups

Let $H, N \leq G$. Their **set product** is

$$HN = \{hn \mid h \in H,\ n \in N\}.$$

Their **join** is

$$H \vee N = \langle H \cup N \rangle,$$

the smallest subgroup of G containing both H and N.

Lemma 0.6.3. Let $H \leq G$ and $N \trianglelefteq G$. Then

$$H \vee N = HN = NH \leq G.$$

If H is also normal in G, then $HN \trianglelefteq G$.

Remark. Without additional hypotheses, the set product of two subgroups need not be a subgroup.

Theorem 0.6.4 (Second Isomorphism Theorem)**.** Let $H \leq G$ and $N \trianglelefteq G$. Then

$$H \cap N \trianglelefteq H, \qquad N \trianglelefteq HN,$$

and there is an isomorphism

$$\boxed{H/(H \cap N) \cong HN/N}$$

given by

$$h(H \cap N) \longmapsto hN.$$

Remark. The homomorphism

$$H \to HN/N, \qquad h \mapsto hN,$$

is surjective and has kernel $H \cap N$. Thus the Second Isomorphism Theorem identifies the image of H under the quotient map $G \to G/N$.

Theorem 0.6.5 (Third Isomorphism Theorem)**.** Let $K, H \trianglelefteq G$ with $K \leq H$. Then

$$H/K \trianglelefteq G/K$$

and there is an isomorphism

$$\boxed{(G/K)/(H/K) \cong G/H}$$

given by

$$(gK)(H/K) \longmapsto gH.$$

Remark. The homomorphism

$$G/K \to G/H, \qquad gK \mapsto gH,$$

is surjective and has kernel H/K. Quotienting first by K and then by H/K therefore has the same effect as quotienting directly by H.

Exercise 0.6.6. Let H, K and L be normal subgroups of G with $H < K < L$. Let $A = G/H, B = K/H, C = L/H$

- Show that B and C are normal subgroups of A, and $B < C$.

  Proof. For $gH \in A, kH \in B$, we have $gHkHg^{-1}H = (gkg^{-1})H = k'H$ for some $k' \in K$, hence $B \trianglelefteq A$. Similarly, we know $C \trianglelefteq A$.
  For $kH \in B$, we know $k \in L$ thus $kH \in C$; $K < L$ means there is $l \in L \setminus K$ thus $lH \in C \setminus B$. It follows that $B < C$. □

- To what factor group of G is $(A/B)/(C/B)$ isomorphic?

  Proof. By Third Isomorphism Theorem we know

  $$(A/B)/(C/B) \cong A/C = (G/H)/(L/H) \cong G/L$$

□

Exercise 0.6.7. Let K and L be normal subgroups of G with $K \vee L = G$, and $K \cap L = \{e\}$. Show that $G/K \cong L$ and $G/L \cong K$.

Proof. Since $K \trianglelefteq G$ and $L \trianglelefteq G$, we know $G = K \vee L = KL$. Then the Second Isomorphism Theorem implies that

$$KL/K = G/K \cong L/(K \cap L) = L/\{e\} \cong L$$

Similar argument shows that $G/L \cong K$. □

Exercise 0.6.8. Let G be a group with subgroups $H^* \leq H \leq G$ and $K^* \leq K \leq G$. Prove that

$$H^*(H \cap K^*) \cap (H \cap K) = (H^* \cap K)(H \cap K^*)$$

Proof. Notice

$$H^*(H \cap K^*) \cap (H \cap K) = ((H^* \cap K) \cup (H^* \setminus K))(H \cap K^*) \cap (H \cap K)$$

Since $H \cap K \leq G$, then for $x \in H \cap K^*, y \in H \cap K$, one has $yx^{-1} \in H \cap K$, it follows that

$$(H^* \setminus K)(H \cap K^*) \cap (H \cap K) = \varnothing$$

then

$$H^*(H \cap K^*) \cap (H \cap K) = (H^* \cap K)(H \cap K^*) \cap (H \cap K) = (H^* \cap K)(H \cap K^*)$$

The second equation is due to $H^* \cap K \leq H \cap K$ and $H \cap K^* \leq H \cap K$. □

1 The Fundamental Group

1.1 Homotopy of Paths

Definition 1.1.1. Homotopic and Nulhomotopic

If f and f' are continuous map of the space X into the space Y, we say f is **homotopic** to f' if there is a continuous map $F : X \times I \to Y$ such that

$$F(x,0) = f(x) \quad \text{and} \quad F(x,1) = f'(x)$$

for each x. The map F is called a **homotopy** between f and f', denoted by $f \simeq f'$. If $f \simeq f'$ and f' is a constant map, we say that f is **nulhomotopic**.

Definition 1.1.2. Path Homotopic

Two paths $f, f' : I \to X$ are said to be **path homotopic** if they have the same initial point x_0 and the same final point x_1, and if there is a continuous map $F : I \times I \to X$ such that

$$F(s,0) = f(s) \quad \text{and} \quad F(s,1) = f'(s)$$
$$F(0,t) = x_0 \quad \text{and} \quad F(1,t) = x_1$$

for each $s, t \in I$. The map F is called a **path homotopy** between f and f', denoted by $f \simeq_p f'$.

Remark. In this case, the relations $\simeq$ and $\simeq_p$ are equivalence relations.

In particular, if f is a path, then its path homotopy equivalence class is denoted by $[f]$.

Remark. Note that generally it's not true that for paths f, g which share the same endpoints, they are homotopic iff they are path homotopic.

The reason is path homotopy requires endpoints to be fixed during the transformation while homotopy does not. As a result, if there is a hole between two endpoints, then path homotopy cannot transform a path on the one side to the other side while homotopy can.

In fact, a path f is always homotopic to the constant map at $f(0)$, thus being nulhomotopic. As a result, we have:

> Paths $f, g : I \to X$ are homotopic if and only if $f(0)$ and $g(0)$ belong to the same path component of X.

Remark. However, let A be any convex subspace of $\mathbb{R}^n$. Then any two paths $f, g : I \to A$ which share the same endpoint are path homomoprhic under the **straight-line homotopy**:

$$F(x,t) = (1-t)f(x) + tg(x)$$

Definition 1.1.3. Product of Paths

Let f be a path in X from x_0 to x_1, g be a path in X from x_1 to x_2. We define the **product** $f * g$ of f and g to be the path h from x_0 to x_2 given by

$$h(s) = \begin{cases} f(2s) & s \in [0, \frac{1}{2}] \\ g(2s-1) & s \in [\frac{1}{2}, 1] \end{cases}$$

It then induces a well-defined operation $*$ on path-homotopy classes defined by

$$\boxed{[f] * [g] = [f * g]}$$

Theorem 1.1.4 (Groupoid Properties)**.** The operation $*$ on path-homotopy classes has the following properties:

- (Associativity) With endpoints match, one has
$$([f] * [g]) * [h] = [f] * ([g] * [h])$$
- (Right and left identities) Let $e_x(s) = x$ be the constant path and f be a path from x_0 to x_1, then
$$[f] * [e_{x_1}] = [f] = [e_{x_0}] * [f]$$
- Let f be a path from x_0 to x_1, $\bar{f}(s) = f(1-s)$ be a path from x_1 to x_0, then
$$[f] * [\bar{f}] = [e_{x_0}] \quad \text{and} \quad [\bar{f}] * [f] = [e_{x_1}]$$

Remark. Let $f, f' : I \to X$ be paths and $k : X \to Y$ be continuous. Then

$$f \simeq_p f' \implies k \circ f \simeq_p k \circ f'$$

Remark. Let $f, g : I \to X$ with $f(1) = g(0)$ and $k : X \to Y$ be continuous. Then

$$k \circ (f * g) = (k \circ f) * (k \circ g)$$

Theorem 1.1.5. Let $f : I \to X$ be a path and

$$0 = a_0 < a_1 < \cdots < a_n = 1$$

Reparametrize each segment as a path on I:

$$f_i(s) = f(a_{i-1} + (a_i - a_{i-1})s)$$

Then

$$[f] = [f_1] * \cdots * [f_n]$$

Exercise 1.1.6. Let $I = [0, 1]$, then for any space X and continuous maps $f, g : X \to I$. We have $f \simeq g$.

Proof. Define $F : X \times I \to I$ by the straight-line homotopy

$$F(x, t) = (1 - t)f(x) + tg(x)$$

Then one can verify that F is a homotopy between f and g. □

Exercise 1.1.7. A space X is said to be **contractible** if the identity map $i_x : X \to X$ is nulhomotopic.

- Show that I and $\mathbb{R}$ are contractible.

Proof. Since both I and $\mathbb{R}$ are both convex subspaces of $\mathbb{R}$. Then $f(x) = 0$ is homotopic to i_x given by the striaght-line homotopy

$$F(x,t) = (1-t)f(x) + ti_x(x)$$

□

- Show that a contractible space is path connected.

  Proof. Let X be a contractible space and $i_x \simeq f$ where $f(x) = y$ for all $x \in X$. Let F be the homotopy from i_x to f. Then given any $z \in X$, the map $g : I \to X$ defined by
  $$g(t) = F(z,t)$$
  is a path from z to y, which implies X is path connected. □

- Let $f, g : X \to Y$ be continuous where Y is contractible, show that $f \simeq g$.

  Proof. Since Y is contractible. Then $i_y \simeq h$ for some $h(y) = y_0$ with homotopy F satisfying $F(y,0) = y$ and $F(y,1) = y_0$. Then $G : X \times I \to Y$ defined by
  $$G(x,t) = F(f(x),t)$$
  is a homotopy from f to $u(x) = y_0$. It follows that $f \simeq g$. □

- Let $f, g : X \to Y$ be continuous where X is contractible and Y is path connected, show that $f \simeq g$.

  Proof. Let $i_x \simeq u$ with $u(x) = x_0$ and homotopy U such that $U(x,0) = x$ and $U(x,1) = x_0$. Define $h_f : X \to Y$ by $h_f(x) = f(x_0)$. We know $f \simeq h_f$ under the homotopy defined by
  $$F(x,t) = f(U(x,t))$$
  Similarly, $g \simeq h_g$. Finally, there is a path $p : I \to Y$ from $f(x_0)$ to $g(x_0)$, which induces the homotopy
  $$P(x,t) = p(t)$$
  implying that $h_f \simeq h_g$. Therefore $f \simeq g$. □

Remark. A space is contractible would imply that space has trivial fundamental group. But S^2 has trivial fundamental group while being not contractible.

1.2 The Fundamental Group

Definition 1.2.1. Fundamental Group

Let X be a space; let $x_0 \in X$. A path in X that begins and ends at x_0 is called a **loop** based on x_0.

The set of path homotopy classes of loops based at x_0, with the operation $*$, form a group, called the **fundamental group** of X relative to the base point x_0, denoted by $\pi_1(X, x_0)$.

Remark. Sometimes this group is called the **first homotopy group** of X.

Example 1.2.2.

- Let X be a convex subset of $\mathbb{R}^n$, then any two loops based at $x_0 \in X$ are path homomorphic under the straight-line homotopy, implying that $\pi_1(X, x_0)$ is the trivial group containing a single element.
- Let $X = S^1$. Then $\pi_1(X, x_0) = \{[f_k] : k \in \mathbb{Z}\} \cong \mathbb{Z}$ where f_0 is the constant loop; f_k represents k clockwise turns for $k > 0$; f_k represents $-k$ counterclockwise turns for $k < 0$.

Definition 1.2.3. $\hat{\alpha}$

Let α be a path in X from x_0 to x_1. We define a map

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$$

by

$$\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha]$$

Theorem 1.2.4. The map $\hat{\alpha}$ is a group isomorphism.

Corollary 1.2.5. If X is path connected, then

$$\pi_1(X, x_0) \cong \pi_1(X, x_1)$$

for any $x_0, x_1 \in X$.

Remark. Let $x_0 \in C \subset X$ where C is a path component of X, then we have

$$\pi_1(C, x_0) = \pi_1(X, x_0)$$

So we usually deal with only path-connected spaces for fundamental groups.

Definition 1.2.6. Simply Connected

A space X is said to be **simply connected** if it is path connected and $\pi_1(X, x_0)$ is the trivial group, denoted $\pi_1(X, x_0) = 0$, for some, and hence for all $x_0 \in X$.

Lemma 1.2.7. The path-connected space X is simply connected if and only if any two paths having the same initial and final points are path homotopic.

Definition 1.2.8. Induced homomorphism

We denote

$$h : (X, x_0) \to (Y, y_0)$$

if $h : X \to Y$ is continuous and $h(x_0) = y_0$. It induces a homomorphism

$$h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$$

defined by

$$h_*([f]) = [h \circ f]$$

often denoted as $(h_{x_0})_*$ to clearify the base point.

Theorem 1.2.9 (Functorial Properties)**.** For continuous maps

$$(X, x_0) \xrightarrow{h} (Y, y_0) \xrightarrow{k} (Z, z_0)$$

we have

$$(k \circ h)_* = k_* \circ h_*$$

For the identity map $i : (X, x_0) \to (X, x_0)$

$$i_* = \mathrm{id}_{\pi_1(X,x_0)}$$

Corollary 1.2.10. If $h : (X, x_0) \to (Y, y_0)$ is a homemorphism of X with Y, then h_* is an isomorphism of $\pi_1(X, x_0)$ with $\pi_1(Y, y_0)$.

Exercise 1.2.11. A subset A of $\mathbb{R}^n$ is said to be **star convex** if for some $a_0 \in A$, all the line segments joining a_0 to other points of A lie in A.

- Find a star convex set that is not convex.

  Proof. The set $\{(x, y) : xy = 0\} \subset \mathbb{R}^2$ is star convex but not convex. □

- Show that if A is star convex, A is simply connected.

  Proof. For all $x \in A$. The line segement $\{\alpha x + (1 - \alpha)a_0 : 0 \leq \alpha \leq 1\}$ is a path from a_0 to x fully contained in A. Therefore, A is path connected.
  Let l be a loop based on a_0. Then F defined by

  $$F(s, t) = ta_0 + (1 - t)l(s)$$

  is a path homotopy from l to the identitfy loop at a_0. It follows that $\pi_1(A, a_0) = 0$. □

Exercise 1.2.12. Let α be a path in X from x_0 to x_1; let β be a path in X from x_1 to x_2. Show that if $\gamma = \alpha * \beta$, then $\hat{\gamma} = \hat{\beta} \circ \hat{\alpha}$

Proof. Let $[f] \in \pi_1(X, x_0)$. Then

$$\hat{\beta}(\hat{\alpha}([f])) = \hat{\beta}([\bar{\alpha}] * [f] * [\alpha]) = [\bar{\beta}] * [\bar{\alpha}] * [f] * [\alpha] * [\beta] = [\bar{\gamma}] * [f] * [\gamma] = \hat{\gamma}([f])$$

□

Exercise 1.2.13. Let X be path-connected and let $x_0, x_1 \in X$. Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1, we have $\hat{\alpha} = \hat{\beta}$.

Proof. $\Longleftarrow$: Let $[f], [g] \in \pi_1(X, x_0)$; let α be a path from x_0 to x_1 and $\beta = f * \alpha$. Then

$$\hat{\beta}([g]) = [\bar{\beta}] * [g] * [\beta] = [\bar{\alpha}] * [\bar{f}] * [g] * [f] * [\alpha]$$

$$\hat{\alpha}([g]) = [\bar{\alpha}] * [g] * [\alpha]$$

It follows that

$$[\bar{f}] * [g] * [f] = [g] \implies [g] * [f] = [f] * [g]$$

$\Longrightarrow$: Let α, β be paths from x_0 to x_1. Let $[f] \in \pi_1(X, x_0)$. Then we have

$$\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha] = [\bar{\alpha}] * ([\beta] * [\bar{\beta}]) * [f] * ([\beta] * [\bar{\beta}]) * [\alpha]$$

We know $\pi_1(X, x_1)$ is also abelian. Then

$$\begin{aligned} &([\bar{\alpha}] * [\beta]), ([\bar{\beta}] * [f] * [\beta]), ([\bar{\beta}] * [\alpha]) \in \pi_1(X, x_1) \\ \implies &\hat{\alpha}([f]) = ([\bar{\alpha}] * [\beta]) * ([\bar{\beta}] * [\alpha]) * ([\bar{\beta}] * [f] * [\beta]) = [\bar{\beta}] * [f] * [\beta] = \hat{\beta}([f]) \end{aligned}$$

□

Exercise 1.2.14. Let $A \subset X$; let $r : X \to A$ be a retraction of X onto A (i.e. r is continuous and $r(a) = a$ for all $a \in A$). If $a_0 \in A$, show that

$$r_* : \pi_1(X, a_0) \to \pi_1(A, a_0)$$

is surjective.

Proof. Let $p : I \to A$ be a loop at a_0. Then $p' : I \to X$ defined by $p' = p$ is also a loop at a_0. Hence $r_*([p']) = [r \circ p'] = [p]$ implying that r_* is surjective. □

Exercise 1.2.15. Show that if X is path-connected, the homomorphism induced by a continuous map is independent of the base point, up to isomorphisms of the groups involved. More precisely, let $h : X \to Y$ be continuous, with $h(x_0) = y_0$ and $h(x_1) = y_1$. Let α be a path in X from x_0 to x_1, let $\beta = h \circ \alpha$. Show that

$$\hat{\beta} \circ (h_{x_0})_* = (h_{x_1})_* \circ \hat{\alpha}$$

Proof. Given $[f] \in \pi_1(X, x_0)$, we have

$$\hat{\beta} \circ (h_{x_0})_*[f] = \hat{\beta}([h(f)]) = [\bar{\beta}][h(f)][\beta]$$

$$(h_{x_1})_* \circ \hat{\alpha}[f] = (h_{x_1})_*([\bar{\alpha}][f][\alpha]) = [h(\bar{\alpha})][h(f)][h\alpha] = [\bar{\beta}][h(f)][\beta]$$

□

Exercise 1.2.16. Let $\langle G, \cdot \rangle$ be a topological group with identity x_0. Let $\Omega(G, x_0)$ denote the set of all loops in G based at x_0. If $f, g \in \Omega(G, x_0)$, then $f \otimes g$ defines a loop by

$$(f \otimes g)(s) = f(s) \cdot g(s)$$

- Show that $\Omega(G, x_0)$ together with the operation $\otimes$ is a group.

Proof. For all $f, g \in \Omega(G, x_0)$, by definition we have $f \otimes g \in \Omega(G, x_0)$.
For all $f, g, h \in \Omega(G, x_0)$, we have

$$(f \otimes g) \otimes h(s) = f \otimes (g \otimes h)(s) = f(s)g(s)h(s)$$

Let $e_{x_0} \in \Omega(G, x_0)$ be the constant loop based at x_0. Then for $f \in \Omega(G, x_0)$, since x_0 is the identity, we have

$$(f \otimes e_{x_0})(s) = f(s)x_0 = f(s) = (e_{x_0} \otimes f)(s)$$

then e_{x_0} is the identity of $\Omega(G, x_0)$.
Let $f \in \Omega(G, x_0)$, then $f' : I \to G$ defined by

$$f'(s) = f^{-1}(s)$$

is trivially the inverse of f. □

- Show that this operation induces group operation $\otimes$ on $\pi_1(G, x_0)$ which is identical to $*$. More precisely, let $f, g \in \Omega(G, x_0)$, then

$$[f * g] = [f \otimes g]$$

Proof. To show that $[f] \otimes [g] = [f \otimes g]$ is well-defined. Let F be the path homotopy between f and f', G be the path homotopy between g and g'. Then H defined by

$$H(s, t) = F(s, t) \cdot G(s, t)$$

can be verified to be the path homotopy between $f \otimes g$ and $f' \otimes g'$.
Therefore, we have

$$[f \otimes g] = [f] \otimes [g] = [f * e_{x_0}] \otimes [e_{x_0} * g] = [(f * e_{x_0}) \otimes (e_{x_0} * g)] = [f * g]$$

□

- Show that $\pi_1(G, x_0)$ is abelian.

Proof. Let $f, g \in \Omega(G, x_0)$. Define g', f' by $g'(s) = g^{-1}(s)$ and $f'(s) = f^{-1}(s)$. Then

$$[\bar{f}][f] = [f'][f] = [e_{x_0}] \implies [\bar{f}] = [f']$$

It follows that

$$[f] * [g] * [\bar{f}] * [\bar{g}] = [(f * g) \otimes (f' * g')] = [e_{x_0}] \implies [f][g] = [g][f]$$

□

1.3 Covering Spaces

Definition 1.3.1. Evenly Covered

Let $p : E \to B$ be a continuous surjective map. The open set $U \subset B$ is **evenly covered** by p if $p^{-1}(U)$ is the union of disjoint open sets of $V_\alpha \subset E$, such that $p|_{V_\alpha}$ is a homeomorphism of V_α onto U. The collection $\{V_\alpha\}$ is a partition of $p^{-1}(U)$ into **slices**.

Remark. If U is evenly covered by p and W is an open set contained in U, then W is also evenly covered by p.

Definition 1.3.2. Covering Map and Covering Space

Let $p : E \to B$ be continuous and surjective. If every point b of B has a neighborhood U that is evenly covered by p, then p is called a **covering map**, and E is said to be a **covering space** of B.

Remark. If $p : E \to B$ is a covering map, then for each $b \in B$, $p^{-1}(\{b\}) \subset E$ has the discrete topology.

Remark. If $p : E \to B$ is a covering map, then p is an open map.

Example 1.3.3. The map $p : R \to S^1$ defined by $p(t) = e^{2\pi it}$ is a covering map.

Definition 1.3.4. Local Homeomorphism

$p : E \to B$ is called a **local homeomorphism** of E with B if each $e \in E$ has a neighborhood U such that $p|_U$ is a homeomorphism of U onto $p(U)$ open in B.

Remark. Every covering map is a local homemorphism.

Example 1.3.5. $p : \mathbb{R}^+ \to S^1$ defined by $p(x) = e^{2\pi ix}$ is a local homeomorphism but not a covering map.

Theorem 1.3.6. Let $p : E \to B$ be a covering map. Let $B_0 \subset B$ and $E_0 = p^{-1}(B_0)$, then the map $p_0 = p|_{E_0}$ is a covering map.

Theorem 1.3.7. Let $p : E \to B$ and $p' : E' \to B'$ be covering maps, then

$$p \times p' : E \times E' \to B \times B'$$

is a covering map.

Exercise 1.3.8. Let Y have the discrete topology. Show that the projection map $\pi_X : X \times Y \to X$ is a covering map.

Proof. Since $\pi_X^{-1}(X) = \cup_{y \in Y} X \times \{y\}$ and each $X \times \{y\}$ is homeomorphic to X. Then we know X is evenly covered by π_X. Continuity and surjectivity hold trivially. It follows that π_X is a covering map. □

Exercise 1.3.9. Let $p : E \to B$ be continuous and surjective. Let $U \subset B$ be open and evenly covered by p. Show that if U is connected, then the partition of $p^{-1}(U)$ into slices is unique.

Proof. We show that each component of $p^{-1}(U)$ is exactly a slice. Since each slice is homemoprhic to the connected space U, thus lying within a single component of $p^{-1}(U)$. If some slice is properly contained in a component, then that compoent is a union of disjoint nonempty open sets, thus being not connected. Therefore, each component is exactly a slice. □

Exercise 1.3.10. Let $p : E \to B$ be a covering map; let B be connected. Show that if $|p^{-1}(\{b_0\})| = k$ for some $b_0 \in B$, then $|p^{-1}(\{b\})| = k$ for all $b \in B$. In such case, E is called a k-fold covering of B.

Proof. For each $b \in B$, let U_b is a neighborhood of b which is evenly covered by p. Notice if $x, y \in U_b$ for some $b \in B$, each slice in $p^{-1}(U_b)$, being homeomorphic to U_b, contains exactly one point of $p^{-1}(\{x\})$ and $p^{-1}(\{y\})$, respectively. It follows that $|p^{-1}(\{x\})| = |p^{-1}(\{y\})|$.

Therefore, let $U = \{b \in B : |p^{-1}(\{b\})| = k\}$, $b_0 \in U$ so it's not empty. We then have

$$\bigcup_{b\in U} U_b \subset U \implies U = \bigcup_{b\in U} U_b$$

implying that U is open. Let x be a limit point of U, let $b \in U_x \cap U$, then

$$b \in U \implies \left|p^{-1}(\{b\})\right| = k \implies \left|p^{-1}(\{x\})\right| = \left|p^{-1}(\{b\})\right| = k \implies x \in U$$

implying that U is closed. Since B is connected, it follows that $U = B$. □

Exercise 1.3.11. Let $q : X \to Y$ and $r : Y \to Z$ be covering maps; let $p = r \circ q$. Show that if $r^{-1}(\{z\})$ is finite for each $z \in Z$, then p is a covering map.

Proof. Let $p = r \circ q$. Since q and r are continuous and surjective, so is p. Fix $z \in Z$, and write

$$r^{-1}(\{z\}) = \{y_1, \ldots, y_n\}.$$

Choose an open neighborhood V of z evenly covered by r. Its slices can be indexed as $V_1, \ldots, V_n$, with $y_i \in V_i$, since each slice contains exactly one point of $r^{-1}(\{z\})$.

For each i, choose an open neighborhood U_i of y_i evenly covered by q. Set

$$W = \bigcap_{i=1}^{n} r(V_i \cap U_i).$$

Since r is open, then $r(V_i \cap U_i)$, implying that W is an open neighborhood of z.

Define

$$W_i = r^{-1}(W) \cap V_i.$$

Then the W_i are pairwise disjoint open sets, and

$$r^{-1}(W) = \bigcup_{i=1}^{n} W_i.$$

Moreover, $r|_{W_i} : W_i \to W$ is a homeomorphism. Since $W \subseteq r(V_i \cap U_i)$ and $r|_{V_i}$ is injective, we have $W_i \subseteq U_i$.

Each W_i is therefore an open subset of an open set evenly covered by q, so it is itself evenly covered by q. Write

$$q^{-1}(W_i) = \bigcup_{\alpha \in J_i} W_{i,\alpha},$$

where the $W_{i,\alpha}$ are pairwise disjoint open sets and each restriction

$$q|_{W_{i,\alpha}} : W_{i,\alpha} \to W_i$$

is a homeomorphism.

Consequently,

$$p^{-1}(W) = \bigcup_{i=1}^{n} q^{-1}(W_i) = \bigcup_{i=1}^{n} \bigcup_{\alpha \in J_i} W_{i,\alpha}.$$

These sets are mutually disjoint: for fixed i they are distinct q-slices, and for distinct i they lie in the preimages of disjoint sets W_i. On each one, the restriction

$$p|_{W_{i,\alpha}} = r|_{W_i} \circ q|_{W_{i,\alpha}}$$

is a homeomorphism onto W.

Thus W is evenly covered by p. Since z was arbitrary, p is a covering map. □

1.4 The Fundamental Group of the Circle

Definition 1.4.1. Lifting

Let $p : E \to B$ be a map. If $f : X \to B$ is continuous, then a **lifting** of f is a map $\tilde{f} : X \to E$ such that $p \circ \tilde{f} = f$.

Lemma 1.4.2. Let $p : E \to B$ be a covering map, let $p(e_0) = b_0$. Any path $f : I \to B$ beginning at b_0 has a unique lifting to a path $\tilde{f}$ in E beginning at e_0.

Lemma 1.4.3. Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$. Let the map $F : I \times I \to B$ be continuous, with $F(0,0) = b_0$. There is a unique lifting of F to a continuous map

$$\tilde{F} : I \times I \to E$$

such that $\tilde{F}(0,0) = e_0$. If F is a path hoomotopy, then $\tilde{F}$ is a path homotopy.

Theorem 1.4.4. Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$. Let f and g be two paths in B from b_0 to b_1; let $\bar{f}$ and $\bar{g}$ be their respective liftings to paths in E beginning at e_0. If f and g are path homotopic, then $\bar{f}$ and $\bar{g}$ are path homotopic as well.

Definition 1.4.5. Lifting Correspondence

Let $p : E \to B$ be a covering map; let $b_0 \in B, e_0 \in E$ such that $p(e_0) = b_0$. Given $[f] \in \pi_1(B, b_0)$, let $\bar{f}$ be the lifting of f to a path in E that begins at e_0. Then we define

$$\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$$

by $\phi([f]) = \bar{f}(1)$, called the **lifting correspondence**.

Theorem 1.4.6. Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$. If E is path connected, then the lifting correspondence

$$\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$$

is surjective. If E is simply connected, it is bijective.

Remark. Injectivity is deduced directly from Lemma 1.2.7.

Theorem 1.4.7. The fundamental group of S^1 is isomorphic to the additive group of integers.

Theorem 1.4.8. Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$.

- The homomorphism $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is a monomorphism.
- Let $H = p_*(\pi_1(E, e_0))$. The lifting correspondence ϕ induces an injective map

  $$\Phi : \pi_1(B, b_0)/H \to p^{-1}(b_0)$$

  which is bijective if E is path-connected.
- If f is a loop in B based at b_0, then $[f] \in H$ if and only if f lifts to a loop in E based at e_0.

Exercise 1.4.9. Let $p : E \to B$ be a covering map, with E path connected. Show that if B is simply connected, then p is a homeomorphism.

Proof. Since $\pi_1(B, b_0)$ is the trivial group. Then by Theorem 1.4.6 we know $|p^{-1}(b_0)| = 1$ for all b_0 being surjectively mapped from a trivial group. It follows that p is injective then bijective. Since p is continuous open map, then p is a homeomorphism. □

1.5 Deformation Retracts and Homotopy Type

Lemma 1.5.1. Let $h, k : (X, x_0) \to (Y, y_0)$. If h and k are homotopic with homotopy $F : X \times I \to Y$, and if $F(x_0, t) = y_0$ for all $t \in I$. Then the homomorphisms h_* and k_* are equal.

Theorem 1.5.2. The inclusion map $j : S^n \to R^{n+1} \setminus \{0\}$ induces an isomorphism of fundamental groups.

Definition 1.5.3. Deformation Retract and Deformation Retraction

Let A be a subspace of X. We say that A is a **deformation retract** of X if there is a continuous map $H : X \times I \to X$ such that $H(x, 0) = x, H(x, 1) \in A$ for all $x \in X$, and $H(a, t) = a$ for all $a \in A, t \in I$. Then H is called a **deformation retraction** of X onto A.

Remark. Let $r : X \to A$ defined by $r(x) = H(x, 1)$ be a retraction. Then H is the homotopy between id_X and $j \circ r$, where $j : A \to X$ is inclusion.

Theorem 1.5.4. Let A be a deformation retract of X; let $x_0 \in A$. Then the inclusion map

$$j : (A, x_0) \to (X, x_0)$$

induces an isomorphism of fundamental groups.

Definition 1.5.5. Homotopy Equivalences and Homotopy Inverse

Let $f : X \to Y$ and $g : Y \to X$ be continuous maps. Suppose that the map $g \circ f : X \to X$ is homotopic to the identity map of X, and the map $f \circ g : Y \to Y$ is homotopic to the identity map of Y. Then the maps f and g are called **homotopy equivalences**, and each is said to be a **homotopy inverse** of the other.

Remark. One can observe that the relation of homotopy equivalence is an equivalence relation.

Definition 1.5.6. Homotopy Type

We call two spaces that are homotopy equivalent to have the same **homotopy type**.

Remark. If A is a deformation retract of X, then A has the same homotopy type as X

Lemma 1.5.7. Let $h, k : X \to Y$ be continuous maps; let $h(x_0) = y_0$ and $k(x_0) = y_1$. If h and k are homotopic, there is a path α in Y from y_0 to y_1 such that $k_* = \hat{a} \circ h_*$. Indeed, if $H : X \times I \to Y$ is the homotopy between h and k, then α is the path $\alpha(t) = H(x_0, t)$.

Corollary 1.5.8. Let $h, k : X \to Y$ be homotopic continuous maps; let $h(x_0) = y_0$ and $k(x_0) = y_1$. If h_* is injective, or surjective, or trivial, so is k_*.

Remark. $\hat{\alpha}$ is a group isomorphism by Theorem 1.2.4

Corollary 1.5.9. Let $h : X \to Y$. If h is nulhomotopic, then h_* is the trivial homomorphism.

Theorem 1.5.10. Let $f : X \to Y$ be continuous; let $f(x_0) = y_0$. If f is a homotopy equivalence, then

$$f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$$

is an isomorphism.

Exercise 1.5.11. Determine for the following spaces whether their fundamental group is trivial, infinite cyclic, or isomorphic to the fundamental group of the figure eight.

Proof.

- The solid torus, $B^2 \times S^1$: Infinite cyclic. It deformation retracts onto its core circle $\{0\} \times S^1$ by contracting each disk cross-section to its centre.
- The torus T with a point removed. Isomorphic to the fundamental group of the figure eight. The punctured square can be deformation retracted to its boundary, then folding opposite edges twices gives a figure eight.
- The cylinder $S^1 \times I$: Infinite cyclic. It deformation retracts onto $S^1 \times \{0\}$.
- The infinite cylinder $S^1 \times \mathbb{R}$. Similarly it can deformation retracts onto $S^1 \times \{0\}$ with deformation retraction $H((u,v),t) = (u,(1-t)v)$.
- $\mathbb{R}^3$ with the nonnegative x, y, z axes deleted. Isomorphic to the fundamental group of the figure eight. Appplying the deformation retraction $c \mapsto \frac{c}{\|c\|}$ gives the unit sphere with three holes. Then expanding a hole gives a bijection to the circle with two holes. Hence it's the same homotopy type as figure eight.

The following subsets of $\mathbb{R}^2$.

- $\{x : \|x\| > 1\}$: Infinite cyclic.
- $\{x : \|x\| \geq 1\}$: Infinite cyclic.
- $\{x : \|x\| < 1\}$: Trivial.
- $\{S^1 \cup \mathbb{R}^+ \times \{0\}\}$: Infinite cyclic. Points in $\mathbb{R}^+ \times \{0\}$ maps to the point $(0,0)$.
- $\{S^1 \cup \mathbb{R}^+ \times \mathbb{R}\}$: Infintie cyclic.
- $S^1 \cup \mathbb{R} \times \{0\}$: Isomorphic to the fundamental group of the figure eight.
- $\mathbb{R}^2 - \mathbb{R}^+ \times \{0\}$: Trivial.

□

Exercise 1.5.12. Let A be a subspace of X; let $j : A \to X$ be the inclusion map, and let $f : X \to A$ be a continuous map. Suppose there is a homotopy $H : X \times I \to X$ between the map $j \circ f$ and the identity map of X.

- Show that if f is a retraction, then j_* is an isomorphism.

  Proof. If f is a retraction, then $f \circ j = \mathrm{id}_A$ implying that j, f are homotopy equivalences, hence j_* is an isomorphism by Theorem 1.5.10 □

- Show that if H maps $A \times I$ into A, then j_* is an isomorphism.

  Proof. We know $f \circ j = f|_A$, then consider $F : A \times I \to A$ defined by $F = H|_{A \times I}$. We have for $a \in A$,

  $$F(a,0) = H(a,0) = j(f(a)) = f(a) \quad \text{and} \quad F(a,1) = H(a,1) = a$$

  implying that F is a homotopy from $f \circ j$ to id_A. Hence j, f are homotopy equivalences and then j_* is an isomorphism. □

- Give an example in which j_* is not an isomorphism.

Proof. Let $A = S^1, X = \mathbb{R}^2$ and f is a constant map. Then $j \circ f$ is a constant map, which is homotopic with id_X. However, j_* is not an isomorphism since the fundamental groups of S^1 and $\mathbb{R}^2$ are not isomorphic. □

1.6 The Fundamental Group of S^n

Theorem 1.6.1. Suppose $X = U \cup V$, where U and V are open sets of X. Suppose that $U \cap V$ is path connected, and that $x_0 \in U \cap V$. Define $i = \mathrm{id}_X|_U, j = \mathrm{id}_X|_V$. Then the images of the induced homomorphisms

$$i_* : \pi_1(U, x_0) \to \pi_1(X, x_0) \quad \text{and} \quad j_* : \pi_1(V, x_0) \to \pi_1(X, x_0)$$

generate $\pi_1(X, x_0)$.

Remark. The theorem tells that any loop in X based at x_0 is path homotopic to a combination of loops in either U or V based at x_0.

Remark. The key point is that any loop in X based at x_0 can be split into paths completely in U or V, using the Lebesgue number lemma.

Corollary 1.6.2. Suppose $X = U \cup V$, where U and V are open sets of X; suppose $U \cap V$ is nonempty and path connected. If U and V are simply connected, then X is simply connected.

Definition 1.6.3. n-sphere

The **n-sphere** S^n is the subspace of $\mathbb{R}^{n+1}$ defined by

$$S^n = \left\{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : x_1^2 + \cdots + x_{n+1}^2 = 1\right\}$$

Definition 1.6.4. Stereographic Projection

Let $p = (0, \ldots, 0, 1) \in S^n \subset \mathbb{R}^{n+1}$, then the function $f : (S^n - p) \to \mathbb{R}^n$ defined by

$$f(x_1, \ldots, x_{n+1}) = \frac{1}{1 - x_{n+1}}(x_1, \ldots, x_n)$$

is a homeomorphism, called the **stereographic projection**.

Therefore, the punctured sphere $S^n - p$ is homeomorphic to $\mathbb{R}^n$.

Theorem 1.6.5. If $n \geq 2$, the n-sphere is simply connected.

Exercise 1.6.6. Let X be the union of two copies of S^2 having a single point in common. What is the fundamental group of X?

Proof. Write $X = S_1 \cup S_2$, where $S_1, S_2 \cong S^2$ and $S_1 \cap S_2 = \{P\}$. We show that every loop based at P is nulhomotopic, which implies that the fundamental group of X is the trivial group.

Choose $q_i \in S_i \setminus \{P\}$ and homeomorphisms

$$\phi_i : U_i := S_i \setminus \{q_i\} \longrightarrow \mathbb{R}^2, \qquad \phi_i(P) = 0.$$

Set $U = U_1 \cup U_2 = X \setminus \{q_1, q_2\}$. Define

$$C(x,t) = \phi_i^{-1}\big((1-t)\phi_i(x)\big) \qquad (x \in U_i,\ t \in I),$$

where $I = [0,1]$. The two formulas agree at P, and U_1, U_2 are closed in U. Hence the pasting lemma shows that $C : U \times I \to U$ is continuous. Moreover,

$$C(x,0) = x, \qquad C(x,1) = P, \qquad C(P,t) = P.$$

Let $f : I \to X$ be a loop based at P, and put

$$F = f^{-1}(\{P\}), \qquad D = f^{-1}(\{q_1, q_2\}).$$

The components of $I \setminus F$ are disjoint open intervals (a,b), with $f(a) = f(b) = P$. By connectedness, the image of each such interval lies entirely in one of $S_1 \setminus \{P\}$ or $S_2 \setminus \{P\}$. Thus $f|_{[a,b]}$ is a loop in one sphere.

Only finitely many of these intervals meet D. Indeed, if $D \neq \varnothing$, then the disjoint compact sets $F, D \subseteq I$ have positive distance δ. Any component (a,b) meeting D consequently has length at least δ, and there can be only finitely many disjoint intervals of that length in I.

Denote these exceptional intervals by $(a_1, b_1), \ldots, (a_m, b_m)$. Since each sphere is simply connected, choose a nulhomotopy

$$H_j : [a_j, b_j] \times I \longrightarrow X$$

of $f|_{[a_j,b_j]}$ that fixes both endpoints at P.

Set

$$K = I \setminus \bigcup_{j=1}^{m} (a_j, b_j).$$

Then K is closed and $f(K) \subseteq U$. Define

$$H(s,t) = \begin{cases} H_j(s,t), & s \in [a_j, b_j], \\ C(f(s),t), & s \in K. \end{cases}$$

These definitions agree on every overlap, since all endpoint values remain P. The sets $K \times I$ and $[a_j, b_j] \times I$ form a finite closed cover of $I \times I$, so the pasting lemma proves that H is continuous. Finally,

$$H(s,0) = f(s), \qquad H(s,1) = P, \qquad H(0,t) = H(1,t) = P.$$

Therefore f is path homotopic to the constant loop. Since X is path connected, its fundamental group is trivial. □

Remark. It should be noted that the union of two simply connected spaces having a point in common is not necessarily simply connected. Consider

$$E = \bigcup_{n=1}^{\infty} C_n \subseteq \mathbb{R}^2 \times \{0\},$$

where C_n has centre $(1/n, 0, 0)$ and radius $1/n$. Let $P = (0, 0, 0)$, $A = (0, 0, 1)$, and define

$$Y = \bigcup_{x \in E} [x, A], \qquad Y' = -Y.$$

Thus Y contains a complete conical side surface over each base circle; the planar base disks are not filled in. Each cone is contractible towards its own apex, and $Y \cap Y' = \{P\}$. Nevertheless, $Y \cup Y'$ is not simply connected.

Let α_n traverse C_n and let β_n traverse its reflected circle, both based at P. The infinite alternating concatenation

$$\gamma = \alpha_1 * \beta_1 * \alpha_2 * \beta_2 * \cdots,$$

traversed on successive intervals accumulating at 1, is continuous because the circles shrink towards P. This loop is not nulhomotopic.

Individually, each α_n can be deformed, with basepoint fixed, into the path PA followed by AP, and then contracted to P. The corresponding statement holds for β_n using the lower apex $B = -A$. However, these individual homotopies cannot simply be performed simultaneously on all the subdivision intervals. At the stage when the upper pieces have become $PA * AP$, there are parameters $s_n \to 1$ whose images are all A, whereas the value at 1 remains P. The resulting intermediate path is therefore discontinuous at 1.

The essential requirement is that shrinking excursions admit shrinking nulhomotopies: if their images approach P, their entire contractions must remain uniformly close to P. For the spheres, the fixed punctured-sphere charts provide precisely this control. The contractions through the cone apices do not: however small a base circle becomes, its chosen contraction still reaches an apex a fixed distance from P.

This explains the failure of the proposed piecewise contraction. The non-nulhomotopy of γ is a separate substantive property of Griffiths' example; it does not follow merely from the failure of that particular contraction.

Exercise 1.6.7. Criticize the following 'proof' that S^2 is simply connected:

> Let f be a loop in S^2 based at x_0. Choose a point p of S^2 not lying in the image of f. Since $S^2 - p$ is homeomorphic with $\mathbb{R}^2$ and $\mathbb{R}^2$ is simply connected, the loop f is then path homotopic to the constant loop.

Proof. Let $f : I \to I^2$ be the Peano space-filling curve such that

$$f(0) = (0, 0) \quad \text{and} \quad f(1) = (0, 1)$$
$$f(I) = I^2$$

Then consider $g : I^2 \to S^2$ be a suejective continuous map defined by

$$g(x, t) = (2\sqrt{x(1-x)}\sin(2\pi t), 2\sqrt{x(1-x)}\cos(2\pi t), 2x - 1)$$

Then $g \circ f : I \to S^2$ is a surjective continuous loop based at $(0, 0, -1)$, which implies that the point p does not necessarily exist. □

Exercise 1.6.8.

- Show that $\mathbb{R}^1$ and $\mathbb{R}^n$ are not homeomorphic if $n > 1$.

  Proof. Since the punctured $\mathbb{R}^1 - x$ is not path connected while any punctured $\mathbb{R}^n - y$ is path connected for $n > 1$, then $\mathbb{R}^1$ is not homeomorphic to $\mathbb{R}^n$ for $n > 1$. □

- Show that $\mathbb{R}^2$ and $\mathbb{R}^n$ are not homeomorphic if $n > 2$.

  Proof. Notice the punctured $\mathbb{R}^2 - x$ can be deformation retracted to S^1 with the fundamental group isomorphic to $\mathbb{Z}$; while the punctured $\mathbb{R}^n - y$ for $n > 2$ can be deformation retracted to S^{n-1} with trivial fundamental group. So the fundamental group of $\mathbb{R}^2 - x$ and $\mathbb{R}^n - y$ is different implies that $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n > 2$. □

Exercise 1.6.9. Suppose $X = U \cup V$, where U and V are open sets of X. Suppose that $U \cap V$ is path connected, and that $x_0 \in U \cap V$. Define $i = \mathrm{id}_X|_U, j = \mathrm{id}_X|_V$.

- What can you say about the fundamental group of X if j_* is the trivial homomorphism? If both i_* and j_* are trivial?

  Proof. If j_* is the trivial homomorphism, then the image of j_* is $\{e\}$, it follows that the fundamental group of X is isomorphic to the image of i_*. If both i_* and j_* are both trivial, then the fundamental group of X is trivial. □

  Remark. It should be noted that the image of i_* may be different from the fundamental group of U. The next subquestion is a typical example.

- Give an example where i_* and j_* are trivial but neither U nor V have trivial fundamental groups.

  Proof. Consider $X = \mathbb{R}^2, p = (0,1) \in \mathbb{R}^2, q = (1,0) \in \mathbb{R}^2$. Then let $U = X - p$ and $V = X - q$ we have both i_* and j_* are trivial but both U and V have fundamental groups isomorphic to $\mathbb{Z}$. □

1.7 Fundamental Groups of Some Surfaces

Theorem 1.7.1.

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

Corollary 1.7.2. The fundamental group of the torus $T = S^1 \times S^1$ is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$.

Definition 1.7.3. The Projective Plane and Projective n-space

The projective plane P^2 is the quotient space obtained from S^2 by identifying each point x of S^2 with its antipodal point $-x$.

The projective n-space is defined similarly, obtained from S^n by identifying each point x with its antipode $-x$.

Theorem 1.7.4. P^2 is a compact surface, and the quotient map $p : S^2 \to P^2$ is a covering map.

Corollary 1.7.5. $\pi_1(P^2, y)$ is a group of order 2.

Remark. Similar for P^n: P^n is a compact surface. $p : S^n \to P^n$ is a covering map. $\pi_1(P^n, y) \cong \mathbb{Z}_2$.

Lemma 1.7.6. The fundamental group of the figure eight is not abelian.

Theorem 1.7.7. The fundamental group of the double torus $T\#T$ is not abelian.

Remark. $T\#T$ can be retracted to the figure eight.

Corollary 1.7.8. $S^2, T, P^2, T\#T$ are topologically distinct.

2 Glossary

Abstract Algebra Foundations

Group

Homomorphism and Isomorphism

Permutation and Symmetric Group

Cycle and Transposition

General Linear Group

Dihedral Group

Subgroup

Cyclic Subgroup and Generator

Greatest Common Divisor

Subgroup Generated by a Subset

Cayley Digraph

Direct Product of Groups

Decomposable and Indecomposable Groups

Coset

Index of a Subgroup

Normal Subgroup

Factor Group

Canonical Projection

Automorphism and Inner Automorphism

Simple Group

Maximal Normal Subgroup

Center of a Group

Commutator and Commutator Subgroup

Abelianization

Product and Join of Subgroups

The Fundamental Group

Homotopic and Nulhomotopic

Path Homotopic

Straight-Line Homotopy

Product of Paths

Contractible

Fundamental Group

$\hat{\alpha}$

Simply Connected

Induced homomorphism

h_*

Evenly Covered

Covering Map and Covering Space

Local Homeomorphism

Lifting

Lifting Correspondence

Deformation Retract and Deformation Retraction

Homotopy Equivalences and Homotopy Inverse

Homotopy Type

n-sphere

Stereographic Projection

The Projective Plane and Projective n-space